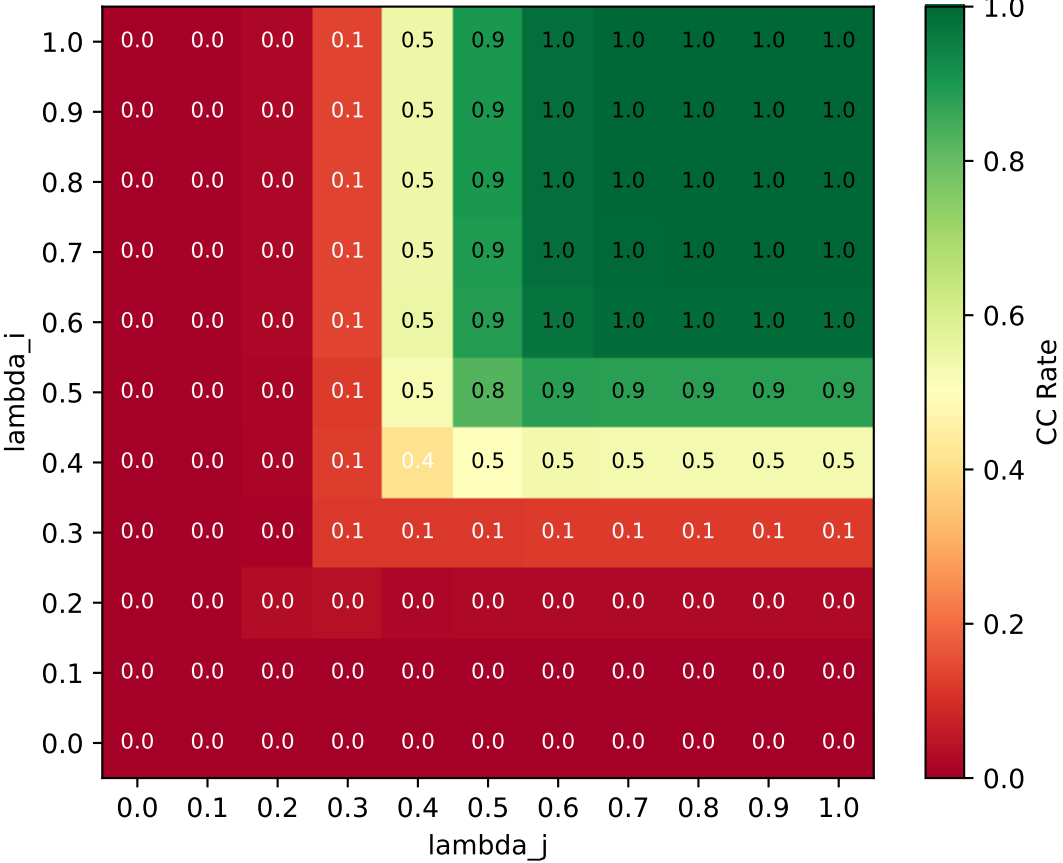
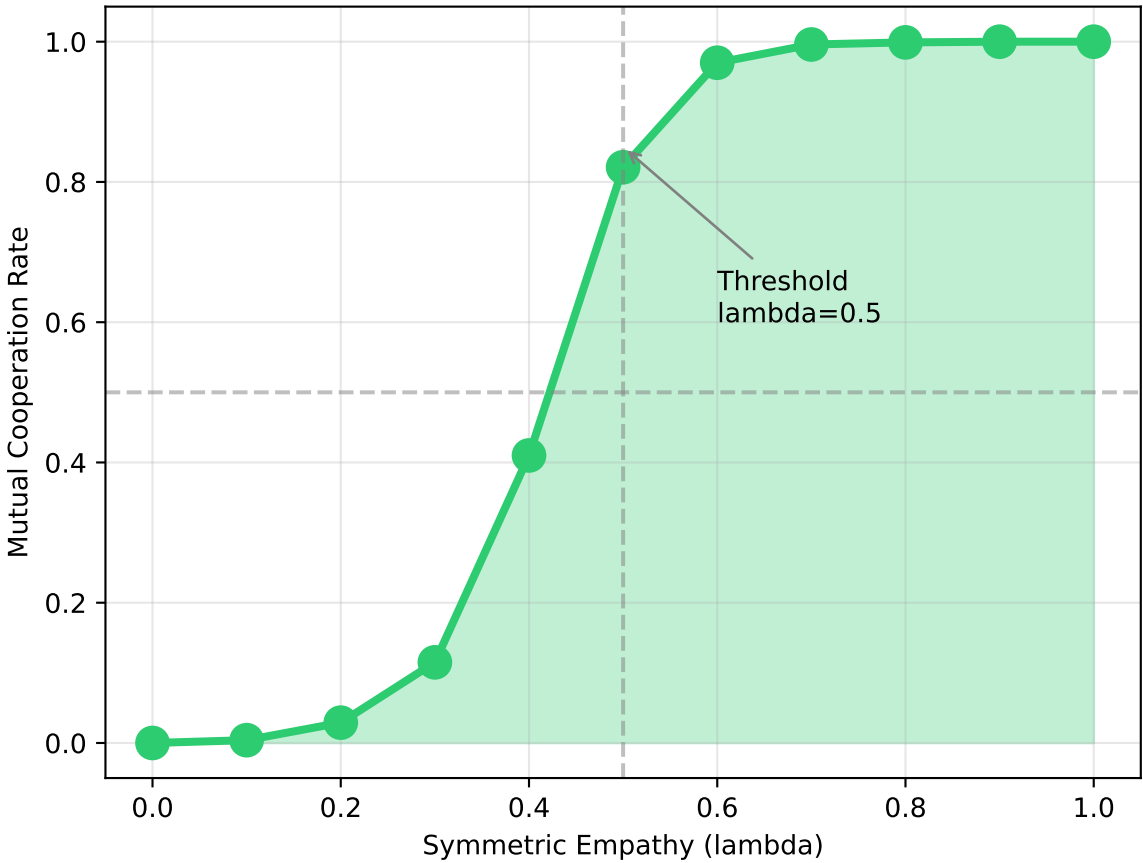


Empathy + ToM in Prisoner's Dilemma: Phase 5 Results

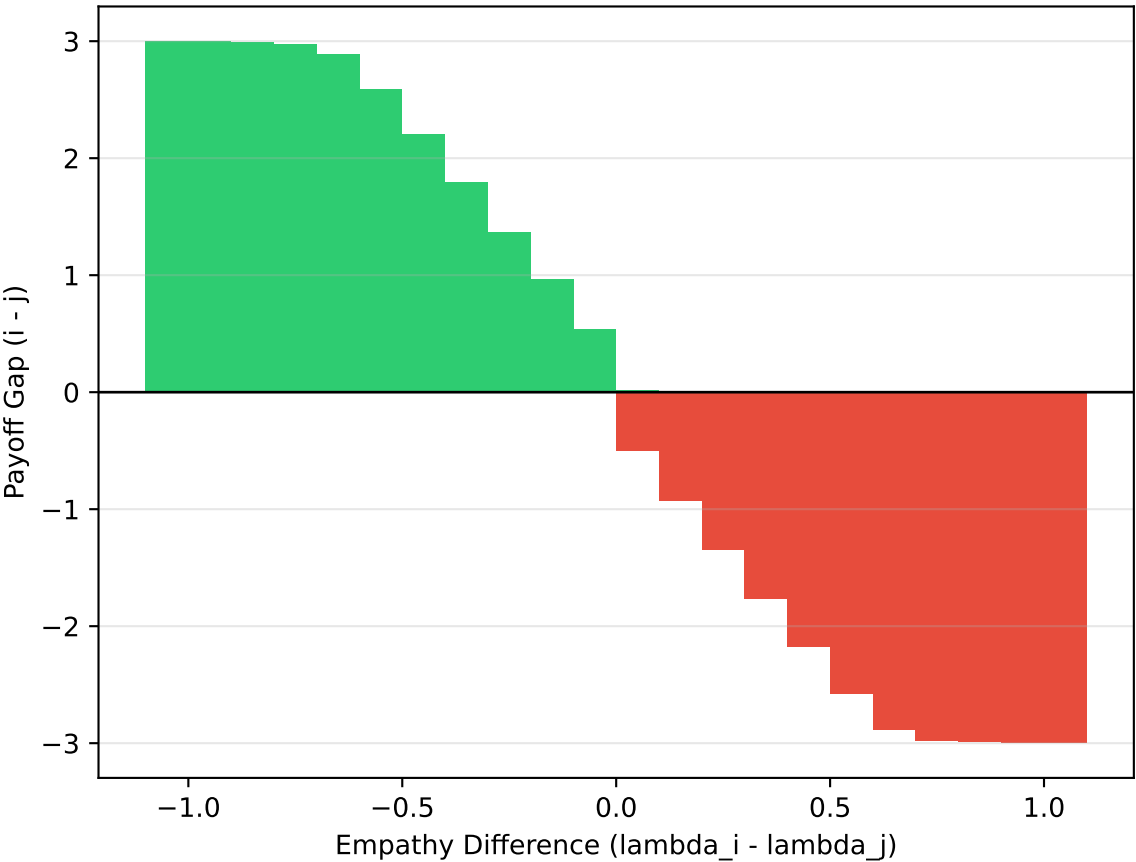
A. Cooperation Rate by Empathy



B. Cooperation Threshold Effect



C. Exploitation: High Empathy Disadvantaged



D. Summary Statistics

D. Key Findings

Total experiments: 2,420

High empathy (both ≥ 0.5):

- Cooperation rate: 95.9%
- N = 720

Low empathy (both ≤ 0.25):

- Cooperation rate: 0.5%
- N = 180

Asymmetric ($|\text{diff}| \geq 0.5$):

- i exploited: 42.5%
- j exploited: 42.8%
- N = 800

Main Result:

Empathy ≥ 0.5 reliably produces mutual cooperation without baking in pro-sociality.